CEHV13 MODULE 04: ENUMERATION

Comprehensive Practical Study Notes & Verification Framework

Enumeration is defined as the phase in ethical hacking where the attacker establishes an active connection to the target host to discover system resources, user accounts, network shares, routing tables, and specific service applications. Unlike passive footprinting, enumeration involves direct queries that generate log entries and traffic alerts.

1. NetBIOS Enumeration

NetBIOS (Network Basic Input/Output System) names identify computers and shared resources on legacy and modern Windows networks. It operates primarily over TCP/UDP ports 137, 138, and 139.

Core Tools & Commands

- `nbtstat` (Windows Native Utility): Evaluates local and remote NetBIOS name caches and tables.
- `nbtstat -a <RemoteName>`: Lists the remote machine's name table given its NetBIOS string.
- `nbtstat -A <IP address>`: Lists the remote machine's name table given its IPv4 address.
- `nbtstat -c`: Displays the local NetBIOS cache containing resolved remote machines and IPs.

Laboratory Evidence Verification:
Executing `nbtstat -c` maps target nodes accurately:

```
NetBIOS Remote Cache Name Table
Name                Type       Host Address      Life [sec]
-----------------------------------------------------
WINDOWS11           <20> UNIQUE   10.10.1.11        326
```

Subsequent execution of `net use` verifies accessible resources:

```
OK          Z:        \WINDOWS11\CEH-Tools     Microsoft Windows Network
```

2. SNMP Enumeration

Simple Network Management Protocol (SNMP) manages network infrastructure. Enumeration attempts to extract system details using read-only passwords known as **community strings** (e.g., `public`).

Automated Information Gathering

```
# snmpwalk -v1 -c public 10.10.1.22
```

This command traverses the Management Information Base (MIB) tree structure to extract critical infrastructure keys (OIDs):

- `iso.3.6.1.2.1.1.1.0` = Hardware description (e.g., `"Hardware: Intel64 Family 6 Model 85 Stepping 7 AT/AT COMPATIBLE - Software: Windows Version 6.3 (Build 2048 Multiprocessor Free)"`).
- `iso.3.6.1.2.1.1.5.0` = System Domain Name / Full FQDN (`"Server2022.CEH.com"`).

3. LDAP Enumeration (Active Directory Mapping)

Lightweight Directory Access Protocol (LDAP) manages hierarchical active directory structures over port 389. Attackers utilize specialized interfaces such as Active Directory Explorer (AD Explorer) to discover administrative associations.

Object Path Hierarchy Analysis

- **Domain Controller Location:** Evaluated using distinguished names: `CN=Domain Controllers,CN=Users,DC=CEH,DC=com` indicating active directories on `10.10.1.22`.
- **User Property Identification:** Discovered attributes map specific target credentials cleanly:
  - **Object:** `CN=Jason M.,CN=Users,DC=CEH,DC=com`
  - **Attribute:** `userPrincipalName`
  - **Value:** `jason@CEH.com`

4. DNS Enumeration & Zone Transfers

DNS enumeration identifies infrastructure layouts by querying public and private name servers. A primary objective is forcing an insecure **Zone Transfer (AXFR)**.

Exploitation & Diagnostics Commands

```
# dig @ns1.bluehost.com www.certifiedhacker.com axfr
```

If misconfigured, this yields the complete DNS map. Secure deployments return `Transfer failed`.

Using the native Windows utility `nslookup`, administrators verify administrative authority via:

```
> set querytype=soa
> certifiedhacker.com
primary name server = ns1.bluehost.com
responsible mail addr = dnsadmin.box5331.bluehost.com
```

5. SMTP & NFS Service Enumeration

SMTP User Harvesting

Mail transfer protocols support discovery commands such as VRFY to validate mailbox names. Attackers use the Nmap Scripting Engine (NSE) to quickly find standard profiles:

```
# nmap -p 25 --script=smtp-open-relay 10.10.1.19
```

This validates whether the system functions as an open relay, indicating vulnerability to unauthorized mail injection.

NFS Service Fingerprinting

Network File System architectures expose storage frameworks directly over unauthenticated ports if improperly locked down. The default standard operation maps to **Port 2049 (TCP/UDP)**.

6. Module Quick Reference Matrix

Enumeration Type	Common Protocols / Ports	Primary Utility	Target Information Discovered
NetBIOS	UDP/TCP 137, 139	`nbtstat / nbtscan`	Local name tables, workstation roles, mapped network shares.
SNMP	UDP 161	`snmpwalk`	OS kernel metrics, detailed hardware architectures, network uptime.
LDAP	TCP/UDP 389	`AD Explorer`	Active Directory organization, account attributes, email links.
DNS	UDP/TCP 53	`dig / nslookup`	Zone tables, MX exchangers, Start of Authority (SOA) managers.
SMTP	TCP 25	`nmap (NSE scripts)`	System usernames (root, admin, guest), open mail relay holes.
NFS	TCP/UDP 2049	`RPCScan`	Exposed network paths, shared directories, file export structures.